← RETURN TO PORTFOLIO

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Jyothika A P

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B.Tech in Computer Science and Engineering · IIIT Kottayam

Kerala, India · Available for AI/ML & Research Internships (2026/2027)

EDUCATION

Indian Institute of Information Technology (IIIT) Kottayam 2023 - 2027
Bachelor of Technology in Computer Science and Engineering

KEY VERIFIED PROJECTS

HIV-ESM2-Enhanced (HEALTH / RESEARCH)

Code

HIV-1 Drug Resistance Prediction with Protein Language Models and Methodological Rigor

Key Result: ~0.9487 — Unbiased nested cross-validation across 18 drugs on full Stanford HIVDB cohort

Technologies: PyTorch, Hugging Face ESM-2 (650M), XGBoost, Optuna, SHAP, Scikit-Learn, BioPython

Shadow-Walk System (SAFETY / SYSTEMS)

Code

Street Vibrancy-Aware Pedestrian Navigation Engine with Real-Time Incident Telemetry

Key Result: Full-Stack Node + Socket.io — Event-driven danger zone emission and dynamic Leaflet routing simulator

Technologies: Node.js, Express, Socket.io, Leaflet.js, OpenStreetMap, JavaScript, HTML5/CSS3

Ullaas (OPPORTUNITY / ACCESS)

Code

Voice-First Multilingual Marketplace for Regional Women Micro-Entrepreneurs

Key Result: Founder-Led Initiative — Conceptual design, conversational flows, and regional accessibility prototype

Technologies: Voice UX Design, React, Python / FastAPI, Web Speech API, Multilingual NLP Concepts

RakshakX (CYBERSECURITY / EDGE AI)

Code

Pre-Action Fraud Interception Agent with On-Device Cross-Channel Threat Correlation

Key Result: NMIT HACKS 2026 — Developed by team InnovateX as Pre-Action Fraud Interception Agent

Technologies: Android, ONNX Runtime, DistilBERT / IndicBERT, Vosk ASR, DNS Sinkholing / VPN API, Python, MITRE ATT&CK

EXPERIENCE & RESEARCH APPOINTMENTS

B.Tech in Computer Science and Engineering — Indian Institute of Information Technology (IIIT) Kottayam

2023 - PRESENT

Deep academic coursework and practical systems development covering Algorithms, Distributed Systems, Software Architecture, Artificial Intelligence, and Quantum Information. Active builder across hackathons and research initiatives.

Research Intern — IIITDM Kurnool

2024

Engaged in academic research exploring computational and algorithmic problem-solving. Deepened exposure to structured experimentation, algorithmic formulation, and scientific literature review.

Lead AI & Security Developer (Team InnovateX) — NMIT HACKS 2026

2026

Collaborated with team InnovateX to conceptualize, architect, and implement RakshakX—a pre-action fraud interception agent featuring on-device NLP, Vosk speech recognition, and MITRE ATT&CK-aligned cross-channel correlation.

Systems & Algorithm Designer — Make Her Hack – CITTIC, CUSAT

2026

Created VELI-FLOW, an estuarine hydrodynamic risk engine predicting tidal-locking waterlogging in Central Kochi using verified astronomical tidal constituents and Dijkstra rerouting.

Founder & Lead Developer — Ullaas Initiative

2024 - PRESENT

Founded Ullaas to build accessible, voice-first technology that empowers regional women micro-entrepreneurs and artisans to participate directly in digital commerce without digital literacy barriers.

Open Source Contributor — GirlScript Summer of Code (GSSoC 2026)

2026

Contributed to open-source software repositories, improving code reliability, submitting verified pull requests, and mentoring fellow contributors on git hygiene and collaborative development.

TECHNICAL COMPETENCIES

AI / ML & RESEARCH:

PyTorch, Hugging Face (ESM-2, DistilBERT), XGBoost & Scikit-Learn, Nested Cross-Validation, SHAP & Explainability, Optuna (Bayesian Hyperopt), ONNX Runtime (Edge ML), Bioinformatics / BioPython

SYSTEMS & WEB:

TypeScript & JavaScript, Next.js & React, Python 3.12 (FastAPI, Flask), Node.js & Socket.io, REST & WebSocket APIs, Tailwind CSS, Leaflet.js & GIS Data (GeoJSON), SQLite & PostgreSQL, Git & Linux (Arch/Ubuntu)

EMERGING & SPECIALIZED:

Google Cirq (Quantum Circuits), AST Static Analysis (Python), MITRE ATT&CK Framework, Android VPN API / DNS Sinkholing, Voice UX / Speech Recognition

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